

Dark Matter as the Metric Shadow of Dead History

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A Flux/MIP Archive Model for Galactic Rotation Curves

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Abstract

The standard interpretation of galactic rotation curves introduces a non-luminous collisionless mass component, while modified-gravity alternatives usually encode the observed excess acceleration through a universal acceleration scale. This paper develops a different interpretation within the Flux/MIP framework: galactic dark-sector behaviour is treated as the metric response to historical registration structure. In this view, baryonic matter is not only a source of mass-energy but also a persistent archive of prior actualization events. We define a global spatial archive depth, introduce a saturating archival-efficiency function, and extend the model with a compact-remnant archive amplifier motivated by white dwarfs, neutron stars, and stellar/rogue black holes as high historical-depth nodes. The resulting v3q model is tested against the official SPARC175 mass-model catalogue. In the present implementation, Newtonian baryons alone give a galaxy-mean MAE of approximately **39.98 km/s**, the v3o spatial archive model gives **27.96 km/s**, and the v3q compact-remnant archive model gives **18.71 km/s**. The model does not yet solve SPARC at MOND-level precision, but it substantially improves over the baryonic baseline and isolates a structured residual pathway associated with old, gas-poor, remnant-rich systems. The central thesis is: **dark matter is the metric shadow of dead history**.

1. Introduction

The galactic rotation-curve problem is usually stated as a missing-mass problem. Observed circular velocities remain too high at large radii compared with the Newtonian prediction from visible baryons. The standard cosmological answer is cold dark matter: an extended collisionless halo surrounding galaxies. However, the tight empirical coupling between baryons and acceleration suggests that the dark-sector signal is not arbitrary. It tracks baryonic structure with remarkable fidelity.

The Flux/MIP programme reframes this coupling. Instead of asking where unseen mass is located, we ask how the metric stores, exports, and responds to historical registration. In this framework, matter is interpreted as bound history: a stable residue of measurement-induced actualization events. Mass is therefore not merely substance in space, but persistent causal record.

The present paper develops the v3q version of this idea for galaxy rotation curves. The model starts with a global archive-depth scalar derived from baryonic surface compression, then adds a saturating efficiency law and a compact-remnant wake amplifier. The SPARC175 dataset provides a first falsification benchmark.

2. Flux/MIP Ontology

The Flux/MIP framework treats physical reality as a system of actualization events. MIP has two associated meanings:

1. **Measurement-Induced Projection:** the transition from unresolved quantum potential into classical registration.
2. **Minimum Intersection Point:** the local condition under which a possibility becomes a persistent event in the metric.

Matter is the stable residue of repeated actualization. The nucleus of an atom is interpreted as a hardened archive of prior registrations, while electron shells form the active interface through which the object exchanges information and entropy with its environment.

The galactic model therefore distinguishes between:

- **spatial compression**, the present baryonic density state;
- **temporal persistence**, the accumulated duration of stable registration;
- **compact-remnant texture**, the hidden population of high-depth archive nodes.

3. Mass as Bound History

In standard dynamics, inertial mass is a scalar resistance to acceleration. In the Flux/MIP interpretation, inertial mass is the work required to re-register an already persistent historical record. This gives a compact statement:

Mass is persistent bound history.

Gravity is then the metric response to gradients and concentrations of this bound historical structure.

This interpretation does not require changing local inertial mass directly. The model instead modifies the effective nonlocal wake produced by the galaxy-level archive. This avoids immediately violating equivalence-principle constraints, because the correction is treated as a metric response rather than a composition-dependent inertial-mass change.

4. Black Holes as Horizon-Locked Historical Archives

Black holes are treated as the limiting case of historical registration. A collapse event occurs when accumulated temporal depth and coherence exceed the local entropy-export capacity:

$$D_t C_{\sigma} / S_{\text{leak}} > \Theta_{\text{BH}}$$

where:

- D_t is temporal depth,
- C_{σ} is coherence/compression,
- S_{leak} is entropy export capacity,
- Θ_{BH} is the horizon-lock threshold.

The event horizon is interpreted as a causal vault: a topological boundary that quarantines saturated historical registration from the surrounding metric. In this language, black holes are not singularities of matter but singularities of accumulated history.

This motivates v3q. Mature galaxies contain compact remnants: white dwarfs, neutron stars, stellar black holes, and potentially rogue black holes. They need not dominate baryonic mass, but they may dominate historical-depth texture.

5. The v3q Master Equations

5.1 Global Spatial Archive Depth

The galaxy is assigned a global baryonic archive depth:

$$D_{\text{spatial}} = (\text{Sigma_bar} / \text{Sigma_ref})^{0.08}$$

with:

$$\text{Sigma_bar} = M_{\text{bar}} / (2 \pi R_{\text{disk}}^2)$$

$$M_{\text{bar}} = M_{\text{star}} + M_{\text{gas}}$$

$$M_{\text{star}} = \text{Upsilon}_{\text{star}} L_{3.6}$$

$$M_{\text{gas}} = 1.33 M_{\text{HI}}$$

$$\text{Sigma_ref} = 4.08 \times 10^9 \text{ Msun kpc}^{-2}$$

$$\text{Upsilon}_{\text{star}} = 0.5$$

5.2 Saturating Archival Efficiency

The v3o causal-slurice efficiency is:

$$\text{eta}(D) = 8.0 D^{-7} / [1 + (D / 0.68)^8]$$

This term prevents runaway archive leakage at high depth.

5.3 Memory Radius

The radial expression scale of the wake is:

$$L_{\text{chi}} = 7.9 R_{\text{disk}} D^{-4.6}$$

Low-depth systems generate broader wakes; high-depth systems compress their archive more tightly.

5.4 Universal Wake Texture

The enclosed historical wake mass is:

$$M_{\text{hist}}(<r) = \text{eta_eff} M_{\text{bar}} x^s / (1+x)^{(s-1)}$$

with:

$$x = r / L_{\text{chi}}$$

$$s = 1.36$$

5.5 Compact-Remnant Archive Proxy

The raw remnant proxy is:

$$R_{\text{rem}} = (1 - f_{\text{gas}}) ((10 - T) / 10)$$

$$f_{\text{gas}} = M_{\text{gas}} / M_{\text{bar}}$$

where T is Hubble type. The term targets old, gas-poor systems.

5.6 Retention Gate

A physically conservative variant requires a sufficiently deep potential to retain and organize compact archive nodes:

$$G_{\text{ret}} = (V_{\text{max}} / V_{\text{ref}})^n / [1 + (V_{\text{max}} / V_{\text{ref}})^n]$$

A useful physical gate explored here is:

$$V_{\text{ref}} = 200 \text{ km/s}$$
$$n = 4$$

The retained remnant archive is:

$$R_{\text{eff}} = R_{\text{rem}} G_{\text{ret}}$$

5.7 Effective Archival Efficiency

The v3q remnant amplifier is:

$$\eta_{\text{eff}} = \eta(D) [1 + \chi_{\text{rem}} R_{\text{eff}}^{\zeta}]$$

with the empirical best ungated values:

$$\chi_{\text{rem}} = 3.0$$
$$\zeta = 0.5$$

The square-root dependence implies sublinear amplification:

$$\eta_{\text{eff}} = \eta(D) [1 + 3 \sqrt{R_{\text{eff}}}]$$

Compact remnants therefore act as a coherence amplifier, not as a linear mass reservoir.

5.8 Total Rotation Curve

The historical acceleration is:

$$g_{\text{hist}}(r) = G M_{\text{hist}}(<r) / r^2$$

The baryonic velocity contribution is computed from SPARC mass-model components:

$$V_{\text{bar}}^2 = |V_{\text{gas}}| V_{\text{gas}} + \text{Upsilon}_{\text{disk}} V_{\text{disk}}^2 + \text{Upsilon}_{\text{bulge}} V_{\text{bul}}^2$$

with:

$$\text{Upsilon}_{\text{disk}} = 0.5$$
$$\text{Upsilon}_{\text{bulge}} = 0.7$$

The final prediction is:

$$V_{\text{Flux}}^2(r) = V_{\text{bar}}^2(r) + G \eta_{\text{eff}} M_{\text{bar}} / r * x^{1.36} / (1+x)^{0.36}$$

6. SPARC175 Validation

The model was evaluated against the SPARC175 rotation-curve dataset using the official global galaxy table and mass-model table. The implementation parses the global table for $L_{3.6}$, R_{disk} , M_{HI} , T , and quality values, and parses the mass-model table for R , V_{obs} , V_{gas} , V_{disk} , and V_{bul} .

6.1 Summary Metrics

Model	Point MAE	Galaxy Mean MAE	Notes
Newtonian baryons	47.01 km/s	39.98 km/s	Fixed baryonic mass-to-light ratios
v3o spatial archive	33.01 km/s	27.96 km/s	Causal-slurice only
v3p-T morphology proxy	28.76 km/s	27.96 km/s	Helpful point-level correction, too blunt galaxy-level
v3q compact-remnant archive	19.59 km/s	18.71 km/s	Best empirical branch
v3q-retention scan best	19.75 km/s	18.92 km/s	Conservative retention-gated branch

6.2 Strongly Improved Systems

Galaxy	T	f_gas	D_spatial	v3o MAE	v3q MAE
UGC02487	0	0.010	0.861	127.98	44.71
NGC6674	3	0.038	0.844	79.80	15.44
NGC3992	4	0.019	0.873	73.03	16.42
UGC02885	5	0.026	0.801	74.77	19.45
NGC5985	3	0.015	0.820	110.22	59.31
NGC2841	3	0.014	0.903	101.94	53.55
NGC5907	5	0.031	0.846	54.44	6.75
UGC12506	6	0.063	0.791	84.76	41.29

6.3 Overcorrection Cases

The first v3q proxy over-amplifies some intermediate systems. These cases motivate the retention-gated form and/or a more refined remnant-population proxy.

Galaxy	T	f_gas	D_spatial	v3o MAE	v3q MAE
NGC4088	4	0.020	0.913	14.19	41.07
NGC6195	3	0.014	0.773	12.36	38.11
NGC4051	4	0.007	0.822	6.47	31.81
NGC4389	4	0.007	0.792	29.91	51.52
NGC5371	4	0.009	0.845	15.95	36.41
UGC11557	8	0.054	0.761	13.51	28.33
NGC4217	3	0.008	0.877	36.04	48.55
F561-1	9	0.096	0.699	5.62	15.54

7. Interpretation

The v3q result suggests that the residual signal in massive early-type systems is not random. The systems that improve most are old, gas-poor, high-luminosity galaxies where compact-remnant archive density should be highest. The square-root scaling indicates that the first compact archive nodes matter disproportionately; after a remnant population exists, additional remnants yield diminishing returns.

This is distinct from MACHO dark matter. The model does not claim that stellar remnants supply all the missing gravitational mass. Instead, compact remnants are treated as high-registration-depth nodes that amplify a nonlocal metric wake.

8. Failure Modes and Falsification Tests

A serious model must be falsifiable. The following tests are direct:

1. **Residual-remnant correlation:** v3q predicts that v3o residuals should correlate with old stellar populations, low gas fraction, and compact-remnant indicators.
2. **Color and stellar-population tests:** External SDSS/WISE/GALEX colours should reduce scatter if temporal persistence is real.
3. **Lensing consistency:** Any archive wake that affects rotation curves must also have a consistent weak-lensing signature.
4. **Bullet Cluster-style systems:** If baryonic archive sources and lensing peaks are fully separated without a historical-lag mechanism, the model fails.
5. **Solar System screening:** The archive wake must vanish in high-coherence local environments.
6. **Independent SPARC reruns:** The model must survive quality cuts, alternate mass-to-light ratios, and independent parsers.

9. Limitations

The current v3q model is not a final replacement for Λ CDM or MOND. It is a structured prototype. Its galaxy-mean MAE remains above MOND-level performance, and the remnant proxy is coarse. Hubble type and gas fraction are not direct measurements of compact-remnant distributions. The retention gate is physically motivated but not yet derived from first principles.

Therefore, the correct empirical claim is modest:

v3q substantially improves over Newtonian baryons and identifies a physically structured residual pathway tied to compact-remnant archive physics.

10. Conclusion

Flux/MIP v3q converts the dark matter question from a missing-substance problem into a missing-history problem. A galaxy is not merely a distribution of visible mass. It is a historical archive containing smooth baryonic depth, temporal persistence, and compact remnant texture.

The present SPARC175 implementation shows a clear improvement from Newtonian baryons to v3o and from v3o to v3q:

39.98 km/s -> 27.96 km/s -> 18.71 km/s

for galaxy-mean MAE.

The strongest conceptual result is:

Dark-sector behaviour is not missing matter alone; it is missing historical registration structure.

Or, in the compact thesis statement:

Dark matter is the metric shadow of dead history.

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